

Localization of Ligand-induced Phosphorylation Sites to Serine Clusters in the C-terminal Domain of the *Dictyostelium* cAMP Receptor, cAR1*

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When *Dictyostelium* cells are stimulated with cyclic adenosine 3',5'-monophosphate (cAMP), the major surface cAMP receptor expressed in early development, cAR1, undergoes a rapid phosphorylation and parallel decrease in electrophoretic mobility which may serve to regulate the activity of this G protein-coupled receptor. Biochemical analyses indicate the electrophoretic mobility shift is caused by phosphorylation of serine residues within the C-terminal cytoplasmic domain. The 18 serines of this domain are grouped in four clusters, designated 1 to 4 (in N- to C-terminal order). Two approaches were taken to determine the distribution of phosphorylation sites among the serine clusters. First, a proteolytic analysis of the C-terminal domain was performed. Second, mutants lacking various combinations of the serine clusters were created by site-directed mutagenesis and their abilities to undergo ligand-induced modification were determined. Both approaches yielded corroborative results consistent with the following model: the stimulus induces the addition of approximately two phosphates to cluster 1 and one to cluster 2; basal phosphorylation occurs predominantly in cluster 3 and to a lesser extent in cluster 2; and cluster 4 is not phosphorylated. The phosphorylation-deficient receptor mutants should be useful for establishing the role of ligand-induced phosphorylation of cAR1 in chemotaxis, cell-cell signaling, and gene expression.

The cell surface receptors for a wide variety of hormones, neurotransmitters, and related extracellular ligands are characterized by a common structure consisting of seven membrane-spanning segments. In general, these receptors transduce signals to effectors by activating intermediary G proteins.¹ Well-studied examples include rhodopsin, β -adrener-

gic receptors, muscarinic acetylcholine receptors, mating factor receptors of yeast, and cAMP receptors of *Dictyostelium* (reviewed in Ref. 1).

Dictyostelium utilizes extracellular cAMP for cell-to-cell communication during a starvation-induced developmental program which transforms a population of unicellular amebas into a multicellular fruiting body. When stimulated by cAMP, cells respond by moving toward the source of cAMP, by relaying the cAMP signal to other cells (termed signaling), and by altering their pattern of gene expression (reviewed in Refs. 2, 3). These responses are mediated by four homologous cAMP receptors (cARs), each transiently expressed at a characteristic stage in development (4, 5).² Targeted gene disruptions have shown that cAR1, which is maximally expressed during the aggregation stage, accounts for most of the cAMP-binding sites and mediates responses to cAMP at this stage (6). cAR2, cAR3, and cAR4 are believed to carry out similar functions at later times in development (7, 8).²

In the signaling response, exogenous cAMP triggers a transient activation of adenylyl cyclase which results in the transient secretion of cAMP. The attenuation of the response, termed adaptation, has been extensively investigated (9). Once cells are adapted to a subsaturating concentration of ligand, they remain responsive to further increments. The magnitude of the elicited response is proportional to the change in receptor occupancy. When a stimulus is removed, cells regain sensitivity (or deadapt) with a half-time of a few minutes (10).

A similar uncoupling process is observed for a variety of G protein-coupled receptors. Studies of β -adrenergic receptors and rhodopsin suggest that ligand-induced phosphorylation plays a fundamental role in this process (reviewed in Ref. 1) and, analogously, the ligand-dependent phosphorylation of cAR1 is correlated, in terms of kinetics and dose-dependence, with adaptation. Dephosphorylation and deadaptation are also coincident (11). These correlations suggest that adaptation of adenylyl cyclase activation and perhaps other cAR1-mediated events require phosphorylation of cAR1.

Previous studies have shown that cAR1 is basally phosphorylated and, upon exposure of cells to cAMP, approximately 4 additional mol of phosphate are rapidly transferred to each mole of receptor (12). Concomitantly, a marked electrophoretic mobility decrease occurs. We demonstrate here that the electrophoretic mobility shift is a consequence of phosphorylation and that basal and ligand-stimulated phosphorylation occurs principally on serine residues within the C-terminal cytoplasmic domain of cAR1. Based on the deduced sequence of cAR1 (4), the 18 serines in the C-terminal cytoplasmic domain can be

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¹ The abbreviations used are: G protein, guanine nucleotide-binding regulatory protein; cAMP, cyclic adenosine 3',5'-monophosphate; cAR, cAMP receptor; CHAPS, 3-[(cholamidopropyl)-dimethylammonio]-1-propanesulfonate; GTP γ S, guanosine 5'-[γ -thio]triphosphate; MES, 2-(N-morpholino)ethanesulfonic acid; NCS, N-chlorosuccinimide;

PAGE, polyacrylamide gel electrophoresis; TAME, N- α -p-tosyl-L-arginine methyl ester; TLCK, N- α -p-tosyl-L-lysine chloromethyl ketone; TPCK, N-tosyl-L-phenylalanine chloromethyl ketone.

² A. Kimmel, personal communication.

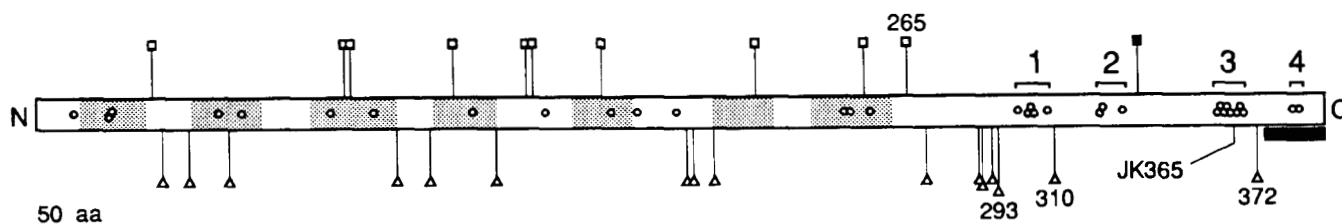


FIG. 1. Schematic view of the cAR1 polypeptide. The positions of serine residues (○) and putative membrane segments (shaded) as proposed by Klein *et al.* (4) are indicated. Brackets numbered 1–4 identify the 4 serine clusters of the C-terminal cytoplasmic domain. Potential NCS cleavage sites including naturally occurring tryptophan residues (□) and a single tryptophan inserted into mutant 335W336 (■) are indicated. Also shown are potential Lys-C cleavage sites (△) and the C terminus of truncated mutant JK365. Residues referred to in the text are numbered. The C terminus-specific antiserum described in the text is directed against the small region identified by the solid bar.

viewed as being organized in four clusters which we have designated, in N- to C-terminal order, as clusters 1 through 4 (Fig. 1). By a combination of proteolytic analysis and site-directed mutagenesis, we have determined the distribution of phosphorylation sites among these serine clusters. The non-phosphorylatable cAR1 mutants which are described here should be useful in studies of the functional significance of cAR1 phosphorylation.

MATERIALS AND METHODS

Mutagenesis of cAR1—cAR1 cDNA 6B (4), subcloned into the *EcoRI* site of bacteriophage M13mp18 DNA, was used as template in oligonucleotide-directed mutagenesis reactions performed as previously described (13). Mutagenic oligonucleotides used are summarized in Table I.

The serines of clusters 1, 2, and 4 were substituted with either alanine or glycine as indicated in Table I. The overlapping oligonucleotides designed to mutate cluster 1 were utilized in two successive rounds of mutagenesis. The 8 serines and single intervening leucine of cluster 3 were deleted. Multiple serine clusters were mutated in reactions including mixtures of appropriate oligonucleotides and either the wild-type or cluster 1-mutated template. Each serine cluster mutant is designated by "cm" followed by 1, 2, 3, and/or 4 indicating which of its serine clusters are mutated. For example, cm1234 is mutated in all four serine clusters. The unmutated, or wild-type, receptor is designated as wt.

Site-directed mutagenesis was also used to insert a tryptophan residue after H³³⁵ in 335W336. The C terminally truncated mutant JK365 was produced by exonuclease III digestion of the cAR1 cDNA. At its C terminus is L³⁶⁵ of cAR1 followed by the vector-encoded pentapeptide -WNSIS.

Transformation of Dictyostelium—Wild-type and mutated cDNAs were ligated into the *EcoRI* site of Bluescript plasmid (Stratagene) which had been modified by the insertion of a *Bgl*II linker into the *EcoRV* site. They were then moved as a *Bam*HI-*Bgl*II fragment to the *Bgl*II site of either of the following G418-selectable expression vectors: pB18, an integrating vector (described in Ref. 14), or pJK1, an extra-chromosomal vector (described in Ref. 15). Expression constructs were introduced into *Dictyostelium discoideum* strain AX3 by electroporation essentially as described by Dynes and Firtel (16). Stably transformed clones, resistant to G418 (20 µg/ml), were screened for cAR1 expression in the vegetative state by immunoblotting.

Cell Growth and Development—AX3 cells were grown in HL5 medium and developed as previously described (17). Development was initiated by washing the cells twice in Mes DB (20 mM MES, pH 6.5, 2 mM MgSO₄, 0.2 mM CaCl₂). Cells at 2 × 10⁷/ml of MES DB were shaken for 4 h with the application of pulses of 50 nM cAMP every 6 min. Transformants were grown in HL-5 containing 20 µg/ml G418 and were used without developing them.

³²P Labeling and cAMP Stimulation of Cells—Developed AX3 cells or vegetative transformants were washed and resuspended at 10⁸ cells/ml in MES DB. Caffeine, which blocks spontaneous cAMP oscillations (17), and 1 mCi of carrier-free ³²P_i (ICN) were added per ml of cell suspension. After shaking for 30 min, half of the labeled cells were stimulated for 5 min with 1 µM cAMP. Dithiothreitol (10 mM) was added with cAMP to inhibit its degradation by extracellular phosphodiesterase. The subscripts "-cAMP" and "+cAMP" modify the names of receptors (wild-type and mutant) isolated from unstimulated and stimulated cells, respectively.

Preparation of Membranes—For experiments utilizing developed

AX3 cells, membranes were prepared immediately following labeling and cAMP stimulation by hypotonic lysis as described by Klein *et al.* (18) except that RB was replaced with ERB (20 mM Tris-HCl, pH 7.5; 5 mM EDTA; 0.5 mM dithiothreitol; 1 mg/ml each of benzamide and 1,10-phenanthroline; 10 µg/ml each of TPCK, TLCK, and TAME; 0.6 µg/ml each of chymostatin and pepstatin A; 0.5 µg/ml antipain; 0.1 µg/ml each of leupeptin and aprotinin; 250 µg/ml *N*-carboxbenzoxyl-L-phenylalanine; 100 µg/ml phenylmethylsulfonyl fluoride). The cAR1 in these membranes was subjected to either SDS solubilization and immunoprecipitation or trypsinization (described below).

Isolation of cAR1 Using CHAPS—Wild-type and mutant forms of cAR1 were enriched by extracting labeled transformed cells three times with ice-cold 1% CHAPS in ERB and 10 min, 12,000 × *g* centrifugations (19). cAR1, which is not solubilized by CHAPS, was then solubilized from the resulting pellet with sample buffer for SDS-PAGE. cAR1 is the predominant phosphoprotein in the resulting preparation (see below). Experiments not shown indicate that this procedure is comparable to immunoprecipitation for isolation of ³²P-labeled cAR1.

cAR1 Immunoprecipitations—Immunoprecipitations were performed as previously described (11). Briefly, membrane-associated cAR1 was first solubilized with sample buffer containing 0.5% SDS and diluted 5-fold with immunoprecipitation buffer to yield a final SDS concentration of 0.1% before the addition of antiserum and protein A-Sepharose. Samples of the soluble C-terminal domain produced by trypsinization of membranes (see below) were immunoprecipitated similarly except they were diluted only 3-fold with immunoprecipitation buffer without introducing SDS. Unless specified otherwise, antiserum raised against intact cAR1 (12) was used.

Production of C Terminus-specific Antiserum—A λgt11 β-galactosidase fusion protein encoding the 18 C-terminal amino acids of cAR1 was gel purified and used to generate rabbit antiserum (4). The ability of a synthetic peptide identical to the 18 C-terminal amino acids to deplete receptor antibodies present in the antiserum demonstrated its specificity for the C terminus of cAR1 (not shown).

Phosphatase Treatment of cAR1—³²P-Labeled cAR1 was immunoprecipitated, and the resulting immune complexes (including protein A-Sepharose) were washed with phosphatase buffer (20 mM HEPES, pH 8.0, 25 mM KCl, 15 mM MgCl₂, 0.05% Triton X-100, and 0.1% SDS) and then resuspended in 100 µl of phosphatase buffer. Calf intestinal alkaline phosphatase (5 units, Boehringer Mannheim) was added where indicated, and samples were incubated at 37 °C for 1 h. After washing once with phosphatase buffer, the treated immunoprecipitates were solubilized with sample buffer for electrophoresis.

Phosphoamino Acid Analysis—³²P-Labeled cAR1 was immunoprecipitated and subjected to SDS-PAGE. Gel regions containing cAR1 were excised, hydrated in 50 mM NH₄HCO₃, and subjected to trypsin proteolysis for 24 h. Material released from the gel fragments was lyophilized, resuspended in 6 N HCl, and hydrolyzed for 2 h at 110 °C under N₂. Hydrolysates were subjected to two dimensional electrophoresis on cellulose plates as previously described (20) followed by autoradiography. Unlabeled phosphoamino acid standards, added to the hydrolysates, were visualized with ninhydrin.

Trypsinization of Membranes—Membranes prepared from labeled AX3 cells were resuspended in ERB lacking TLCK, phenylmethylsulfonyl fluoride, leupeptin, and aprotinin and were treated with trypsin (Worthington) at room temperature with occasional mixing. Unless otherwise indicated, digestions were performed with 100 µg/ml trypsin for 30 min and terminated by the addition of soybean trypsin inhibitor (Worthington) to a final concentration of 150 µg/ml. Samples were centrifuged at 10,000 revolutions/min for 15 min and the supernatants recovered. As described in the text, this treatment liberates a soluble

TABLE I
Oligonucleotides used for site-directed mutagenesis of cAR1

Sequence ^a	Mutation effected ^b	Serine Cluster Affected ^c
CAATAACAATCCAGCACCATAAT	Ser ²⁹⁹ →Ala	1
CACCATATGGTGCAGGTCTGGTAC	Ser ³⁰² →Gly, Ser ³⁰³ →Ala, Ser ³⁰⁴ →Gly	1
CGTGGTACTGCTGGTAAACTA	Ser ³⁰⁸ →Ala	1
GTGCAATGTGGTCCGATATG	Ser ³²⁴ →Gly, Ser ³²⁵ →Ala	2
GAGCAATGTGCTTTGGAAAG	Ser ³³¹ →Ala	2
AATGATGAAGGTΔGATGAAGAAAAAC	Deletes Ser ³⁶⁰ through Ser ³⁶⁸	3
AATATTCAAATTGCCACTGCAACAAATGG	Ser ³⁸³ →Ala, Ser ³⁸⁵ →Ala	4
GGAAAGACATtgGCCAAATATGG	Inserts Trp after His ³³⁵	NA

^a Although antisense oligonucleotides were used, their sense sequences are shown in 5'→3' orientation. Underlined nucleotides represent substitutions, lowercase identifies inserted nucleotides, and the symbol Δ marks the position of deleted nucleotides.

^b Affected residues are numbered relative to the N-terminal methionine of cAR1. Arrows denote substitutions.

^c Serine clusters defined as in Fig. 1. NA, not applicable.

C-terminal receptor fragment.

Affinity Purification of the C-terminal Tryptic Fragment—cAR1 C terminus-specific antibodies were covalently coupled to protein A-Sepharose CL4B (Pharmacia LKB Biotechnology Inc.) as described previously (21). Up to 10 volumes of supernatant from trypsinized membranes were applied to a column of the resin which was then extensively washed with 0.2 M triethanolamine, pH 8.2, 0.5% sodium deoxycholate followed by 0.2 M triethanolamine, pH 8.2, 0.5 M NaCl and finally eluted with 0.2 M triethanolamine, pH 12. Fractions containing the C-terminal fragment were identified by SDS-PAGE, pooled, and dialyzed against 5 mM EDTA, 20 mM Tris, pH 8.0. The immunoaffinity matrix quantitatively removed the C-terminal fragment from the extract and yields were typically 50–90%.

Electrophoresis—Unless specified otherwise, SDS-PAGE was performed according to Laemmli (22). Some receptor digests were analyzed by "low molecular weight" SDS-PAGE on 18% gels as previously described (23). In contrast to standard gels, the separating gels of these have a pH of 9.3 and contain 10% glycerol and a acrylamide/bis-acrylamide ratio of 200:1.

Cleavage by N-Chlorosuccinimide (NCS)—Wild-type and mutant cAR1 were excised from gels and treated with NCS as previously described (24). The treated gel pieces were briefly washed, dried, and inserted into the wells of a second gel. NCS treatment of Lys-C digestion products was performed by lyophilizing and reconstituting them in 50% acetic acid containing 5 mg/ml NCS. Following a 30-min incubation, samples were concentrated and acetic acid removed by evaporation prior to SDS-PAGE.

Endoproteinase Lys-C Digestion—The affinity-purified C-terminal fragment was digested with endoproteinase Lys-C (Boehringer Mannheim; 100–200 ng added/100 μl sample) at 37 °C for the indicated times. Alternatively, wild-type and mutant forms of intact cAR1 from transformants were excised from gels. The rehydrated gel pieces were mixed overnight with 2 volumes of 30 mM Tris-HCl, pH 8.0, 7.5 mM EDTA, and 0.075% SDS containing 1 μg/ml Lys-C. The eluates, typically containing 70% of the total radioactivity, were lyophilized and reconstituted in sample buffer for electrophoresis.

Quantitation of Mutant Receptor Phosphorylation—In order to quantitate the amount of cAR1 on immunoblots, a standard curve relating cAR1-bound ¹²⁵I and the amount of cAR1 protein was constructed as follows. Increasing amounts of a single ³²P-labeled wt_{cAMP} sample were subjected to standard SDS-PAGE and electrophoretically transferred to a polyvinylidene difluoride membrane (Millipore). The membrane was then probed with C terminus-specific antiserum followed by ¹²⁵I-protein A. Using an autoradiographic image as a guide, each region of the membrane containing cAR1 was excised, and the associated ³²P and ¹²⁵I were quantitated by Cerenkov and gamma counting. Background radioactivities were subtracted, and ¹²⁵I measurements were corrected for ³²P which could be detected by gamma counting. The amount of cAR1 was taken as the measured ³²P counts/min (i.e. the amount of the cAR1 protein bound to the polyvinylidene difluoride membrane is proportional to the cAR1-associated ³²P measured). The relationship between ¹²⁵I and ³²P (the measure of cAR1 protein) was nearly linear.

Samples of wt_{cAMP}, cm2, and cm13, labeled with ³²P in parallel with the wt_{cAMP} sample, were subjected to this analysis. cAR1 protein, expressed in arbitrary units, was determined for these samples from their measured ¹²⁵I using the standard curve. The ¹²⁵I measurements used fell within the range of the standard curve. The C terminus-specific antiserum used would be expected to recognize these receptors equally well since the epitope does not include serine clusters 1, 2, or 3.

RESULTS

Relationship of Phosphorylation to Electrophoretic Mobility Shift—The expression of cAR1 is developmentally regulated, reaching maximal levels after 4–6 h of development (12). As previously reported and shown in Fig. 2A, stimulation of developed AX3 cells with 1 μM cAMP induced a rapid phosphorylation and concomitant decrease in electrophoretic mobility of cAR1 (lanes 1 and 2). Treatment of cAR1 extracted from basal and cAMP-stimulated cells (referred to as cAR1_{cAMP} and cAR1_{+cAMP}) with alkaline phosphatase resulted in the loss of ³²P_i from both forms (lanes 3 and 4). Unlabeled cAR1 samples received the same treatments and their electrophoretic mobilities were analyzed by immunoblotting (Fig. 2B). Following *in vitro* dephosphorylation, cAR1_{+cAMP} comigrated with cAR1_{cAMP} while dephosphorylation of cAR1_{cAMP} did not alter its mobility.

We next subjected cAR1_{cAMP} and cAR1_{+cAMP} to phosphoamino acid analysis in order to identify the amino acids modified. As shown in Fig. 2C, cAR1_{cAMP} contained predominantly phosphoserine while cAR1_{+cAMP} contained a trace amount of phosphothreonine in addition to phosphoserine. No phosphotyrosine was detected in either form. Nearly all of the 5-fold increase in phosphate incorporation induced by the stimulus occurred on serine residues. Taken together, these results demonstrate that the observed transition in electrophoretic mobility of cAR1 is a direct result of stimulus-induced addition of phosphates to serine residues.

Localization of the Major Site of Basal and Ligand-induced Phosphorylation—According to the proposed topological model of cAR1 (4), there are 20 cytoplasmically disposed serine residues; 2 are within the loop which joins the fifth and sixth transmembrane segments, and the remaining 18 are in the C-terminal cytoplasmic domain. Tao and Klein (25) proposed that a 23-kDa phosphopeptide, generated by chymotrypsin treatment of cells, represents the C-terminal half of cAR1 and contains these 20 serines. Using controlled proteolysis, we obtained evidence that the sites of phosphorylation are localized to the large C-terminal domain. Membranes containing ³²P_i-labeled cAR1_{+cAMP} were digested with increasing amounts of trypsin (Fig. 3A). At concentrations above 1 μg/ml, the proteolytic treatment released a soluble phosphorylated polypeptide of 19 kDa (lanes 7 and 8). Over 90% of the cAR1 present in the membranes could be digested, and 25–35% of the released radioactivity appeared in the 19-kDa fragment. This polypeptide was effectively immunoprecipitated with a C terminus-specific antiserum (Fig. 3A). When the tryptic fragments released from cAR1_{cAMP} and cAR1_{+cAMP} were compared (Fig. 3B, lanes 1 and 3), the stimulus was observed to induce a slight increase in apparent molecular weight and a 3–5-fold increase in phosphate incorporation similar to what is observed for the intact

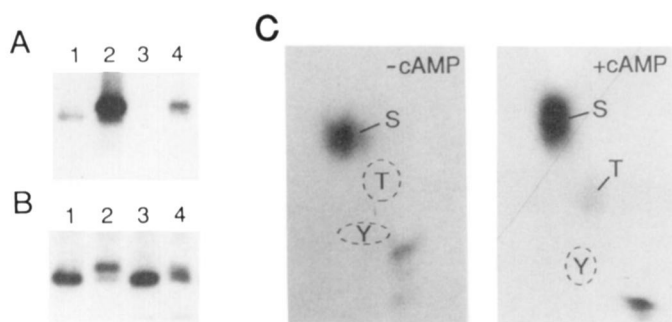


FIG. 2. *In vitro* dephosphorylation of receptors and phosphoamino acid analysis. A, ^{32}P -labeled cAR1_{-cAMP} (odd lanes) and cAR1_{+cAMP} (even lanes) from developed AX3 cells were treated with alkaline phosphatase as described under "Materials and Methods." Samples were incubated without (lanes 1 and 2) or with phosphatase (lanes 3 and 4) and subjected to SDS-PAGE followed by autoradiography. B, unlabeled samples were treated identically and analyzed by immunoblotting with cAR1 antiserum (11). C, ^{32}P -labeled samples of cAR1_{-cAMP} and cAR1_{+cAMP} from AX3 cells were analyzed for phosphoamino acid composition as described under "Materials and Methods." S, T, and Y mark the positions of phosphoserine, phosphothreonine, and phosphotyrosine standards.

receptor. The solubility and large apparent molecular weight of these fragments suggest that proteolysis occurred in the cytoplasmic domain at a site near the membrane, possibly at any of 4 lysine residues between positions 287 and 293 (Fig. 1). Cleavage at any of these sites would be expected to yield a ~15-kDa fragment.

For the study of plasmid-encoded cAR1 mutants, we chose to express these and the wild-type receptor under the control of an actin promoter in growth stage cells which express little endogenous cAR1. The plasmid-encoded wild-type receptor is referred to as wt. We previously reported that wt exhibits basal and cAMP-stimulated phosphorylation and an electrophoretic mobility shift which are indistinguishable from those of endogenous cAR1 (14). As shown in Fig. 3B (lanes 5 and 7), the radioactive tryptic fragments derived from wt_{-cAMP} and wt_{+cAMP} also could not be distinguished from those derived from cAR1.

Based on the trypsin proteolysis results, we predicted that all of the incorporated radioactivity would be present on a single NCS cleavage product of 127 residues since there is only 1 tryptophan in the C-terminal domain (Trp²⁶⁵, see Fig. 1). As shown in Fig. 3C, the major phosphopeptide generated upon NCS cleavage of wt_{+cAMP} has an apparent size of 27 kDa. The same treatment of wt_{-cAMP} yielded a less radioactive phosphopeptide with a slightly faster mobility (Fig. 3C). These data suggest that the major NCS phosphopeptide is basally phosphorylated and, like the intact receptor, undergoes a shift in electrophoretic mobility upon further phosphorylation in response to cAMP.

Although these NCS fragments migrate more slowly than expected, several lines of evidence verify their C-terminal origin. First, they could be immunoprecipitated using C terminus-specific antiserum (not shown). Second, C-terminal truncation of the receptor (*mutant JK365*; refer to Fig. 1) resulted in smaller NCS phosphopeptides of 22 and 23 kDa (Fig. 3C). Third, when a tryptophan residue was inserted by site-directed mutagenesis after His³³⁵ within the C-terminal domain in mutant 335W336 (see Fig. 1), the NCS treatment produced a smaller 17-kDa phosphopeptide, indicating that the novel NCS site had been introduced into the phosphorylated region (Fig. 3C). These observations support the view that both basal and stimulated phosphorylation sites are primarily located in the C-terminal domain.

Phosphopeptide Mapping—As an initial approach to further

localize the phosphorylated residues, we digested cAR1 samples with endoproteinase Lys-C which cleaves specifically at lysine residues. A complete digestion with this enzyme is expected to yield only three serine-containing peptides from the C-terminal cytoplasmic domain: a 17-residue peptide (Asn²⁹⁴-Lys³¹⁰) containing serine cluster 1, a 62-residue peptide (Thr³¹¹-Lys³⁷²) which includes both clusters 2 and 3, and a cluster 4-containing, 20-residue peptide (Gln³⁷³-Asn³⁹²) derived from the C terminus (see Fig. 1). The ^{32}P -labeled tryptic fragments from cAR1_{-cAMP} and cAR1_{+cAMP} were completely digested with Lys-C (Fig. 3B). The cAR1_{-cAMP} sample yielded an 11-kDa fragment (lane 2), and the cAR1_{+cAMP} sample gave rise to a 10-kDa fragment and a dye-front fragment (lane 4). Identical results were obtained when the tryptic fragments of wt_{-cAMP} and wt_{+cAMP} were subjected to complete Lys-C digestion (lanes 6 and 8).

In order to identify the Lys-C fragments, we carefully examined the time course of Lys-C digestion of the C-terminal domain from cAR1_{+cAMP} (Fig. 4). After 30 min of digestion (lane 4), the domain was cleaved to three labeled fragments, a 16 kDa band containing about half of the radioactivity, a minor 10 kDa band, and a fragment migrating at the dye front containing the remainder of the radiolabel. A small amount of apparently undigested C-terminal domain also remained. With increasing time of incubation, radiolabel was gradually lost from the 16 kDa band and quantitatively appeared in the 10-kDa fragment (lanes 5–7). There was no further increase in the amount of radiolabel at the dye front. Since multiple additions of Lys-C over 6 h gave the same result as a single addition (lanes 7 and 8), the digestion was presumably complete. Omission of Lys-C demonstrated that proteolysis was not due to instability of the receptor preparation (lane 3). The 16-kDa intermediate was immunoprecipitated by the C terminus-specific antiserum, whereas the dye-front fragment and the 10-kDa peptide failed to be immunoprecipitated (not shown).

Our interpretation of these data is as follows. First, cleavage at Lys³¹⁰ occurs producing the small cluster 1-containing peptide which migrates at the dye front and the 16-kDa intermediate which contains clusters 2, 3, and 4 and the C terminus. Subsequent cleavage of the 16-kDa peptide at Lys³⁷² yields the 10-kDa fragment containing clusters 2 and 3 and the small cluster 4-containing peptide. The latter is probably not labeled since the radioactivity is quantitatively converted from the 16- to the 10-kDa fragment, and there are no immunoprecipitable ^{32}P -labeled products in the complete digest. Thus, the small cAR1_{+cAMP}-derived fragment migrating at the dye front, tentatively identified as containing cluster 1, represents a major site of stimulus-induced phosphorylation. Presumably, the 11 and 10 kDa bands from cAR1_{-cAMP} and cAR1_{+cAMP}, respectively, represent the same polypeptide which contains clusters 2 and 3 and much of the basal phosphorylation. In addition, this polypeptide exhibits some cAMP-dependent phosphorylation which likely accounts for the observed electrophoretic mobility increase.

Localization of Phosphorylation Sites to Specific Serine Clusters—To independently confirm and extend our understanding of the distribution of phosphorylation, we expressed mutant receptors in which various combinations of serine clusters were eliminated by site-directed mutagenesis and analyzed these for their capacities to undergo ligand-induced modifications. As demonstrated by immunoblotting (Fig. 5A), elimination of clusters 2, 3, or 4 in cm234 and cm3 (and cm2 and cm4, not shown) does not impair the cAMP-dependent transition in electrophoretic mobility. Thus, none of the serines of clusters 2, 3, or 4 is necessary for this property. In contrast, elimination of the cluster 1 serines in cm1, cm13, and cm1234 results in receptors which do not exhibit the typical

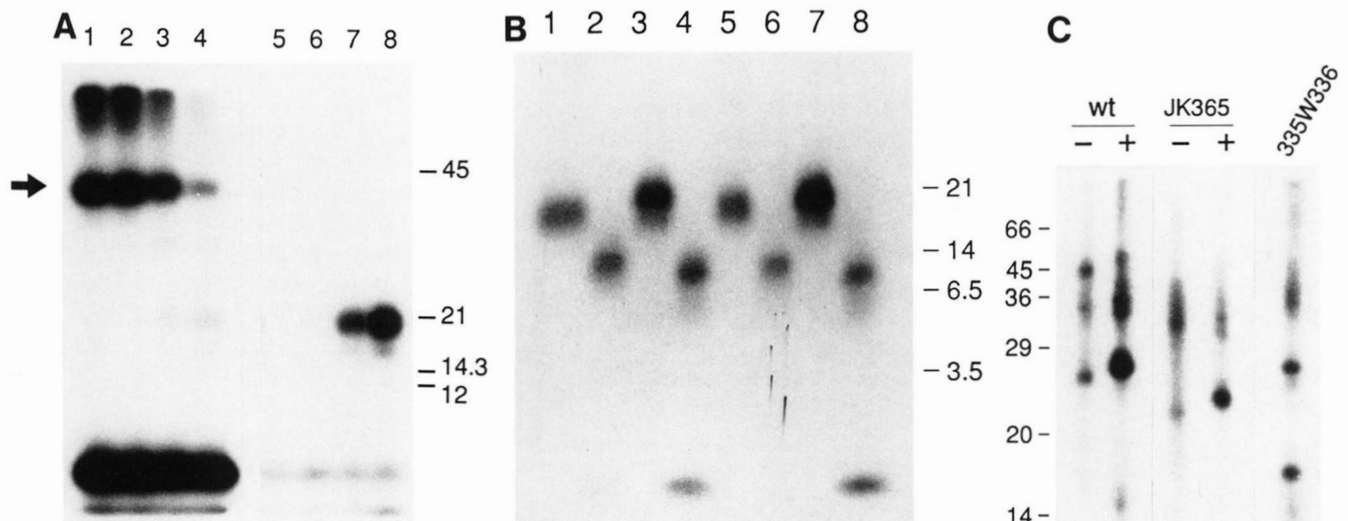


FIG. 3. Localization of phosphorylation sites to the C-terminal domain. A, membranes prepared from ^{32}P -labeled, developed AX3 cells containing cAR1_{+cAMP} were treated for 45 min with 0.1 (lanes 1 and 5), 1 (lanes 2 and 6), 10 (lanes 3 and 7), and 100 (lanes 4 and 8) $\mu\text{g}/\text{ml}$ trypsin and subjected to centrifugation. The resulting pellets (lanes 1–4) and supernatants (lanes 5–8) were subjected immunoprecipitation with C terminus-specific antiserum followed by SDS-PAGE and autoradiography. An arrow marks the position of intact cAR1. The higher molecular weight band is most likely aggregated cAR1. Labeled material near the dye front in lanes 1–4 is presumed to be phospholipid. B, affinity-purified, ^{32}P -labeled C-terminal domains derived from cAR1_{-cAMP} (lanes 1 and 2) and cAR1_{+cAMP} (lanes 3 and 4) were prepared from developed AX3 cells. Similarly, those derived from wt_{-cAMP} (lanes 5 and 6) and wt_{+cAMP} (lanes 7 and 8) were prepared from vegetative cAR1-expressing transformants. Samples in odd lanes received no further treatment; those in even lanes were digested for 6 h with Lys-C. C, NCS digests of ^{32}P -labeled wt_{-cAMP}, wt_{+cAMP}, JK365_{-cAMP}, JK365_{+cAMP}, and 335W336_{+cAMP} are shown analyzed by SDS-PAGE followed by autoradiography. Under the conditions used, NCS is known to yield incomplete digests (24) which probably accounts for many of the high molecular weight phosphopeptides present in the digests shown. This is strongly suggested by the JK365 digests in which their mobilities are systematically increased compared with those in wild-type digests. Similarly, the prominent 27 kDa band derived from 335W336 probably results from failed cleavage at the site of the inserted tryptophan. In this and subsequent figures, molecular masses of standards are shown in kDa.

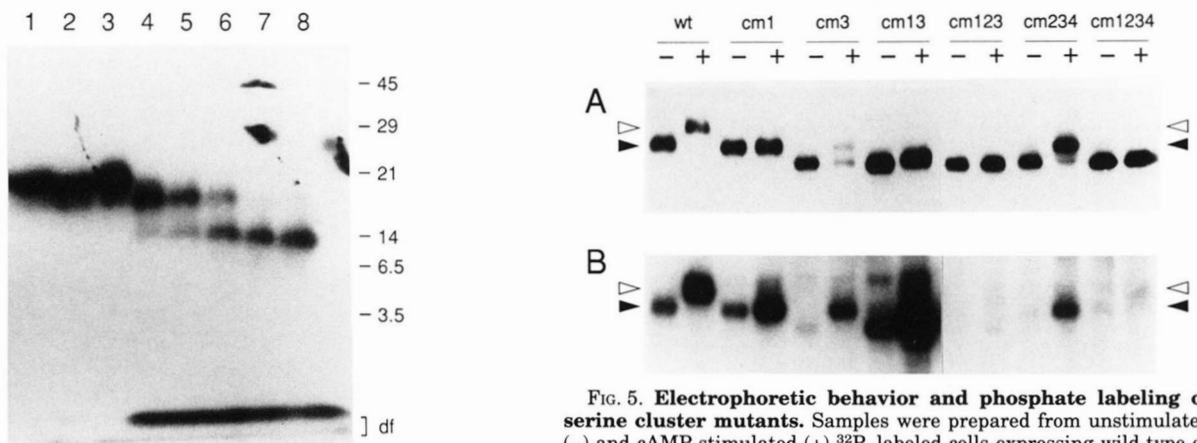


FIG. 4. Time course of Lys-C digestion of the C-terminal domain. Affinity-purified, ^{32}P -labeled C-terminal domain from cAR1_{+cAMP} was subjected to Lys-C digestion followed by SDS-PAGE and autoradiography. Lane 1, untreated; lane 2, 6 h with sample buffer and Lys-C added; lane 3, 6 h with no added enzyme; lanes 4–7, 30 min, 60 min, 2 h, and 6 h with Lys-C; lane 8, 6 h with Lys-C added at 0 h, 2 h, and 4 h. df, dye front.

ligand-induced electrophoretic shift indicating that phosphorylation within cluster 1 underlies the major transition in mobility. A slight cAMP-induced decrease in mobility is evident in cm1 and cm13, but not cm123 and cm1234. This may be due to cAMP-stimulated phosphorylation of cluster 2; however, we did not investigate this further.

The mutant receptors deficient in 1 or more serine clusters were also analyzed for their abilities to undergo basal and cAMP-stimulated phosphorylation. Our observations, illustrated in Fig. 5B, are summarized as follows: (i) cm123 (and cm1234) does not incorporate detectable amounts of ^{32}P providing further confirmation that the C-terminal domain is the principal site of both basal and cAMP-stimulated phosphoryla-

FIG. 5. Electrophoretic behavior and phosphate labeling of serine cluster mutants. Samples were prepared from unstimulated (–) and cAMP-stimulated (+) ^{32}P -labeled cells expressing wild-type or mutated cAR1 as indicated, subjected to SDS-PAGE, and electrophoretically transferred to a nitrocellulose membrane. A, cAR1 immunoblot. The membrane was probed with cAR1-specific antiserum followed by chemiluminescent detection (Amersham Corp.). In the example shown, cm3 shifted incompletely presumably due to inadequate exposure to cAMP; its shift is complete in other experiments not shown. B, cAR1-associated ^{32}P . The nitrocellulose membrane shown in panel A was subsequently rinsed in buffer containing azide to eliminate residual chemiluminescence and subjected to autoradiography to detect ^{32}P . cm123, cm234, and cm1234 were analyzed in a separate experiment and, therefore, cannot be directly compared with the other samples shown. Arrowheads mark the positions of wt_{-cAMP} (filled) and wt_{+cAMP} (unfilled). Note that mutants in which cluster 3 is deleted have increased mobilities.

tion and indicating that neither of the 2 serines of cluster 4 is phosphorylated to a significant extent. (ii) cm234 exhibits only ligand-dependent phosphorylation, indicating that cluster 1 contains a site (or sites) for stimulated, but not basal, phosphorylation. (iii) cm13, which has only clusters 2 and 4, exhibits both basal and ligand-stimulated phosphorylation. Since cluster 4 does not appear to undergo phosphorylation, this indicates that cluster 2 contains sites for both basal and stimulated

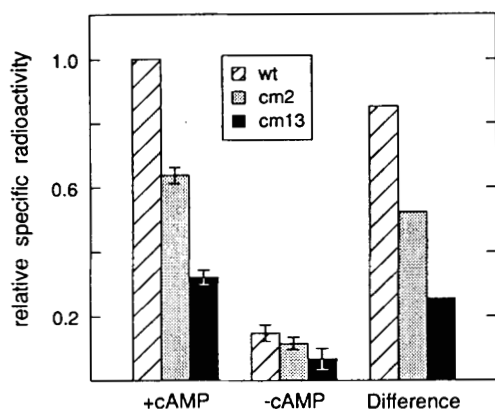


FIG. 6. Quantitation of mutant receptor phosphorylation. cAR1-associated ^{32}P and cAR1 protein were quantitated as described under "Materials and Methods" for wild-type and selected mutant receptors. For each sample, relative specific radioactivity was calculated as the ratio of cAR1 ^{32}P and cAR1 protein and normalized to that of wt_{+cAMP}. Two determinations made for each sample (indicated by brackets) were averaged. The differences between unstimulated and cAMP-stimulated values obtained for each receptor type are also shown and reflect the ligand-dependent component of phosphorylation.

phosphorylation. (iv) Substitution of cluster 2 (mutant cm2) fails to eliminate basal phosphorylation indicating that it occurs on sites outside of cluster 2 (not shown). Sites within cluster 3 likely account for this basal phosphorylation since, as discussed above, cm234 and cm123 argue that sites within clusters 1 and 4, respectively, are not basally phosphorylated. (v) cm124 exhibits basal phosphorylation but little, if any, cAMP-stimulated phosphorylation (not shown). This result and analysis of a Lys-C digest of cm2 (see below) suggest that phosphorylation of cluster 3 is largely basal. This is consistent with the large cAMP-stimulated increase in phosphorylation of cm3 (Fig. 5B).

Quantitation of the Extent of Phosphorylation of Serine Clusters—Since clusters 1, 2, and 3 appeared to account for most of the cAR1 phosphorylation, we decided to compare the extents of phosphorylation of wt, cm2, and cm13. ^{32}P -Labeled wild-type and mutant receptors, derived from cAMP-stimulated or unstimulated cells, were immunoblotted, permitting a quantitative measurement of phosphorylation as described under "Materials and Methods." The ratio of cAR1-associated ^{32}P to cAR1 protein was determined for each sample, and these ratios were normalized to that of wt_{+cAMP} as depicted in Fig. 6. cm2_{+cAMP} and cm13_{+cAMP} yielded values of 0.64 and 0.32, respectively, consistent with their having reduced amounts of phosphorylation compared with wt_{+cAMP}. Significantly, the sum of these values is approximately 1. This is consistent with the supposition that clusters 1, 2, and 3 account for the large majority of phosphorylation. It also suggests that the clusters are phosphorylated independently of one another.

The values determined for cAMP-stimulated samples reflect the sum of both basal and cAMP-dependent incorporation of ^{32}P . The ligand-dependent component of phosphorylation (*i.e.* the difference between stimulated and basal values) for each mutant is shown in Fig. 6. These values indicate that roughly one-third of ligand-dependent phosphorylation is found within cluster 2 and that the remaining two-thirds of ligand-dependent phosphorylation occur in clusters 1 and/or 3. Since cluster 3 appears not to be a target of ligand-dependent phosphorylation, two-thirds of ligand-dependent phosphorylation most likely occur within cluster 1. These values could be accommodated if, upon binding cAMP, cAR1 is phosphorylated on 2 serines in cluster 1 and 1 serine in cluster 2. This supposition agrees reasonably well with our previously reported determi-

nation that roughly four phosphates are added in response to ligand (12).

The extents of phosphorylation of wt_{-cAMP} and cm2_{-cAMP} are nearly equal, suggesting that cluster 2 accounts for only a minor fraction of basal phosphorylation. The low extent of phosphorylation of cm13_{-cAMP} suggests that much of the basal phosphorylation resides within cluster 3. The brief labeling conditions employed may underestimate the amount of basal phosphorylation since it requires the turnover of cAR1-bound phosphate.

Analysis of the Serine Cluster Mutants by Peptide Mapping—In order to confirm the proposed identities of the Lys-C phosphopeptides, the ^{32}P -labeled serine cluster mutants were digested with Lys-C and analyzed by SDS-PAGE. For these experiments, intact receptors, rather than C-terminal tryptic fragments, were digested with Lys-C, and the products were analyzed using an electrophoretic system designed to resolve low molecular weight peptides. For comparison of the standard and low molecular weight gel systems, digests of wt_{-cAMP} and wt_{+cAMP} were resolved by each. The 11- and 10-kDa peptides seen previously, believed to contain clusters 2 and 3, appear as large phosphopeptides ranging in apparent size from roughly 6 to 15 kDa on the standard gel (Fig. 7A, left). The material running with the tracking dye, believed to contain cluster 1, again was only present in the wt_{+cAMP} digest. When the low molecular weight system was used (Fig. 7A, right), all of the phosphorylated species had higher apparent molecular weights, and the material that migrated with the tracking dye on standard gels was resolved into several discrete species with apparent sizes of ~4 kDa and below.

When digests of mutants were analyzed, several observations verified that the large fragments encompass both clusters 2 and 3. First, when a tryptophan residue was inserted between clusters 2 and 3, these Lys-C phosphopeptides were rendered susceptible to cleavage by NCS (Fig. 7B). Second, these species are absent from Lys-C digests of cm234 which lacks both clusters 2 and 3 (Fig. 7C). Third, as noted previously and demonstrated in Fig. 7, the large Lys-C phosphopeptides from wt exhibit an increase in ^{32}P incorporation and a slight increase in electrophoretic mobility upon stimulation with cAMP. Those peptides derived from cm2, however, exhibit neither of these properties (Fig. 7C). This observation also confirms that cluster 3 is not a major site of cAMP-stimulated phosphorylation. Lastly, the large Lys-C phosphopeptides derived from cm13 are smaller as would be expected due to the deletion of cluster 3 (Fig. 7C).

The small Lys-C phosphopeptides were confirmed to contain cluster 1 since these were absent from digests of cm1_{+cAMP} which lacks cluster 1 (Fig. 7C). (The origin of the minor ~4.4-kDa species in this digest is unknown.) Conversely, the collection of small phosphopeptides was unaltered by the elimination of clusters 2, 3, and 4 in cm234 (Fig. 7C). The heterogeneity of this cluster 1-containing peptide may reflect different patterns of phosphorylation or alternative cleavage products resulting from the utilization of different sites by Lys-C among the 4 closely spaced lysine residues at its N-terminal end (see Fig. 1).

DISCUSSION

Stimulation of cells with cAMP triggers an increase in phosphorylation and a decrease in the electrophoretic mobility of cAR1. The kinetics and dose dependence of these two processes are similar. Treatment of cAR1_{+cAMP} with alkaline phosphatase resulted in the removal of phosphate and a consequent decrease in apparent size so that it comigrated with cAR1_{-cAMP}, demonstrating the electrophoretic mobility shift produced *in vivo* upon stimulation is a direct result of phosphate addition. We have previously shown that once the transition from 40 to

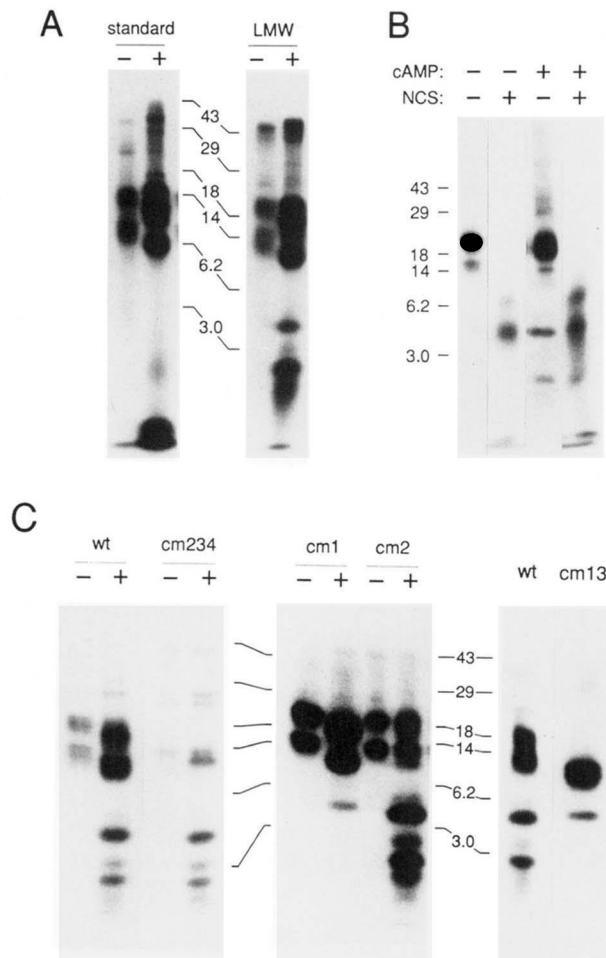


FIG. 7. Lys-C phosphopeptides derived from wild-type and mutant forms of cAR1. ^{32}P -labeled receptors purified from unstimulated (-) and cAMP-stimulated (+) cells were digested with Lys-C. Unless specified otherwise, the resulting phosphopeptides were subjected to low molecular weight SDS-PAGE followed by autoradiography. **A**, digests of wt_{-cAMP} and wt_{+cAMP} were analyzed by standard SDS-PAGE (left) and low molecular weight SDS-PAGE (right). **B**, Lys-C digests of 335W336 are shown with or without subsequent NCS treatment as indicated. **C**, Lys-C digests of wild-type cAR1, cm234, cm1, cm2, and cm13 are shown. At the right, only cAMP-stimulated samples of wt and cm13 are shown.

43 kDa has occurred, this form can acquire additional phosphates without a further increase in apparent molecular mass (11). Thus, not all of these phosphates are required to produce the apparent size change.

The hydrophilic C-terminal domain of cAR1 was revealed as the major site of both basal and ligand-induced phosphorylation. Upon trypsinization of membranes isolated from unstimulated and stimulated cells labeled *in vivo* with $^{32}\text{P}_i$, soluble 18- and 19-kDa phosphopeptides were released which could be immunoprecipitated with a cAR1 C terminus-specific antiserum. Their large size, solubility, and immunoreactivity suggest that these polypeptides represent much of the C-terminal cytoplasmic domain as defined in the topological model of cAR1 (4). Consistent with this was the finding that the radiolabel appeared in a large NCS fragment which was made smaller by C-terminal truncation of cAR1 or insertion of a novel NCS cleavage site within the C-terminal domain.

The 18 serines in the C-terminal domain are organized in four clusters as illustrated in Fig. 1. To determine how the multiple sites of phosphorylation are distributed among these four clusters, we took two independent approaches. The first was a proteolytic analysis in which the trypsin-released do-

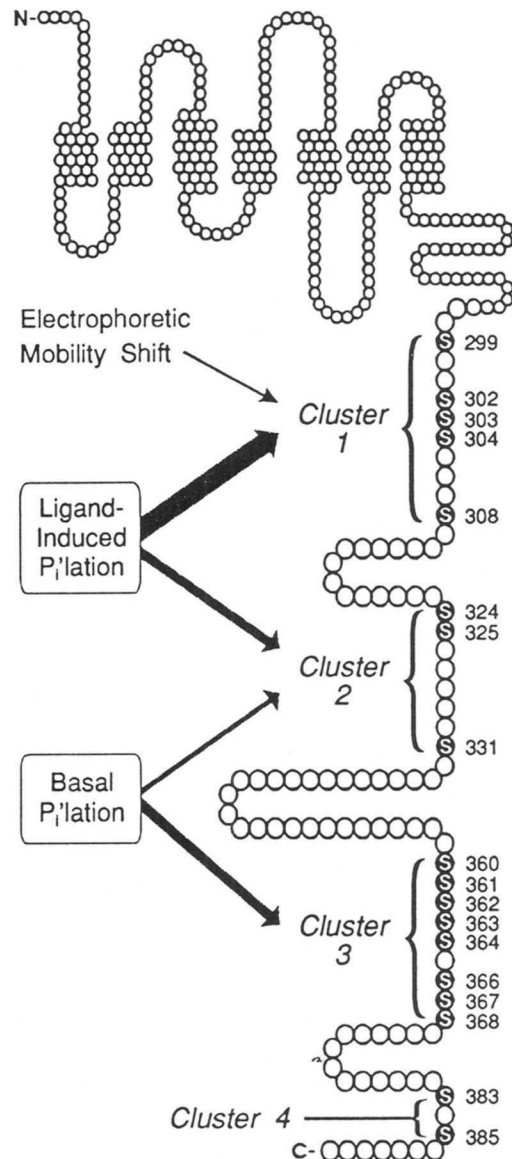


FIG. 8. Model of cAR1 phosphorylation. Serines of the C-terminal domain, numbered relative to the N-terminal methionine, and their assignments to one of four clusters in the text are depicted to the left of the C-terminal domain. The relative distribution of ligand-induced and basal phosphorylation is roughly indicated by arrow width. The electrophoretic mobility shift results from ligand-induced phosphorylation within cluster 1.

main was further digested with Lys-C. Characterization of the phosphopeptide products strongly suggested that serine cluster 1 was only phosphorylated in response to ligand and that cluster 2 and/or cluster 3 contained the sites of basal phosphorylation and some ligand-induced phosphorylation. The second approach, which verified and extended these conclusions, utilized site-directed mutants in which serine clusters in various combinations were eliminated. These were examined for their ability to be phosphorylated and undergo the ligand-induced electrophoretic mobility shift. In addition, the extents of phosphorylation of selected mutants were quantitated. Taken together, our findings are consistent with the model depicted in Fig. 8. Serine cluster 1 contains only stimulus-dependent sites, including that required for the ligand-induced electrophoretic mobility shift. Cluster 2 contains basal and stimulus-dependent sites. The stimulus induces the addition of approximately two phosphates to cluster 1 and one to cluster 2. Basal phos-

phorylation occurs predominately in cluster 3 and to a lesser extent in cluster 2.

Serine cluster 1 has the attributes of a regulatory region. This cluster, among the four, undergoes the most dramatic change in response to stimulation, having no basal sites and acquiring perhaps two modifications. Two of the serines in cluster 1, Ser²⁹⁹ and Ser³⁰⁴, are conserved in all four cARs (7, 8).² We have previously reported that cAR2 and cAR3 also exhibit cAMP-induced phosphorylation (7). It is possible that the conserved serines of cluster 1 are phosphorylated and mediate similar functions in all of the cARs. The importance of cluster 2 cannot be dismissed. Two of the serines in this cluster are conserved in cAR2 and could mediate properties specific to these two receptor subtypes.

The ligand-induced electrophoretic mobility shift caused by phosphorylation within cluster 1 possibly reflects an ionic interaction, which persists during electrophoresis, between a phosphoserine in cluster 1 and an adjacent basic residue. The residues involved might be Ser³⁰⁴ and Arg³⁰⁵. In cAR2 which also exhibits a ligand-induced mobility shift (26), Ser³⁰³ (which corresponds to Ser³⁰⁴ in cAR1) is flanked on either side by basic residues. In contrast, cAR3 does not undergo an apparent mobility shift and lacks a corresponding basic residue (7). It remains to be determined whether the electrophoretic mobility shift has a functional correlate *in vivo*.

In a wide variety of G protein-linked receptor systems such as those mediating vision and hormone action in vertebrates and pheromone sensing in yeast, desensitization phenomena prevent cells from continuously responding to persistent stimulation. Processes which contribute to desensitization include uncoupling of the signaling pathways, loss of ligand binding sites without receptor loss, and receptor degradation (reviewed in Ref. 1). All of these processes occur in *Dictyostelium* (27). The process we have termed adaptation, which reversibly uncouples cAR1 from several effectors including adenylyl cyclase, regulates the periodic waves of cAMP which direct the cells to move to aggregation centers. Adaptation may also enhance the capacity of these cells to sense shallow gradients of cAMP (28). In each of these systems, evidence suggests that ligand-induced phosphorylation of the C-terminal cytoplasmic domains or third intracellular loops of the receptor is involved in the desensitization (reviewed in Ref. 1). These observations suggest a remarkable evolutionary conservation of this regulatory mechanism.

Ligand-induced phosphorylation of G protein-linked receptors is mediated by cAMP-dependent protein kinase in some instances and, more generally, by members of a recently defined family of receptor kinases. The family includes rhodopsin kinase (29), two closely related β -adrenergic receptor kinases (Refs. 30, 31), and two homologues in *Drosophila* (32). The receptor kinases appear to be activated when they bind to ligand-occupied receptors (33, 34). Synthetic peptides have implicated interactions between β -adrenergic receptor kinase and the first and third cytoplasmic loops of the β -adrenergic receptor (35). In addition to activating the kinase, these interactions may serve to position it and, thereby, play a role in phosphorylation site determination. Other studies with peptide substrates suggest that both β -adrenergic receptor kinase and rhodopsin kinase prefer to phosphorylate sites with nearby acidic residues (36).

Presumably there is a β -adrenergic receptor kinase-like receptor kinase in *Dictyostelium*. Protein kinase A probably does not phosphorylate cAR1 since it does not contain evident protein kinase A target sequences, and the reaction is insensitive to inhibitors of cAMP synthesis (12) and disruption of the ad-

enylyl cyclase gene.³ We previously demonstrated a cAR1 phosphorylating activity in the CHAPS-insoluble fraction of cells which was enhanced by pretreatment of cells with cAMP (37). More recently, Meier and Klein (38) reported a cAMP receptor kinase activity in plasma membrane preparations which could be stimulated by cAMP *in vitro*. The cAR1 kinase is evidently expressed in both growing and aggregation stage cells since the phosphorylation of overexpressed cAR1 in growing cells is indistinguishable from that in developed cells. The same kinase may also phosphorylate cAR2, cAR3, and cAR4; the highly conserved intracellular loops of the four cARs (7, 8)² might facilitate their recognition by a single kinase. Our new understanding of the distribution of cAMP-induced phosphorylation sites of cAR1 should allow us to develop an accurate *in vitro* assay for the cAR kinase utilizing appropriate synthetic peptides.

In the models put forth to explain uncoupling in the mammalian systems, auxiliary proteins of the arrestin family bind to the phosphorylated receptors and interfere with functional interaction between the ligand-activated receptor and target G proteins (reviewed in Ref. 1). These events essentially neutralize the excited receptor, allowing the G protein to return to its basal state. In contrast, evidence suggests the mechanism of adaptation differs in *Dictyostelium*. It appears that cAR1 signals to downstream components which are the actual sites of adaptation. We and others have shown that GTP γ S, presumably acting independently of receptors, stimulates adenylyl cyclase activity in cell lysates. However, in lysates from adapted cells, GTP γ S can no longer stimulate the adenylyl cyclase (39, 40). This adaptation does not occur in *car1*⁻ cells, demonstrating that the receptor is required for this process to occur (41).

We are currently attempting to express the phosphorylation-deficient cAR1 mutants described here in *car1*⁻ cells which will enable us to determine the importance of cAR1 phosphorylation to adaptation and the developmental program. Our preliminary evidence indicates that *car1*⁻ cells expressing cm1234 develop abnormally which presumably reflects the physiological importance of cAR1 phosphorylation. The collection of mutants described here will enable us to determine which of the serine clusters are responsible for the phenotype observed for cm1234. Important sites within these clusters can be more precisely localized by substitution of the individual serines and their phosphorylation confirmed by Edman degradation of isolated peptides.

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